Tutorial 1

Risk, Prospect Theory, and Bayesian updating

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How this tutorial works

- Five blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 1a, 1b, 2, 3a, 3b, 4a, 4b.

Plan

1. Experimental method: treatments, designs, control
2. Expected utility, risk attitudes, the Holt–Laury menu
3. The Allais paradox and Prospect Theory
4. Bayes' rule, base rates, representativeness
5. Probability matching

Sample *oral* questions at the end.

PART 1

Experimental method

Experimental method

Lecture 1a – "Treatment", "Between- and within-subject designs", "Advantages of experiments"

Treatment = fully specified set of procedures: instructions, incentives, rules of play. Vary **one thing at a time**.

	each subject sees	the threat to identification
Between-subject	one treatment	composition of the two groups
Within-subject	two or more	order / sequence effects

Control: clear incentives, unbiased terminology, no deception.

Confound: T1 = small + real, T2 = large + hypothetical $\Rightarrow$ two factors move together, neither is identified.

Exercise 1.1

Design

Consider a design in which each participant takes part in one session only, and each session is run under a single set of instructions: half of the sessions use real payoffs, the other half hypothetical payoffs.

Is this a between-subject or a within-subject design?

a. Within-subject
b. Between-subject

Exercise 1.2

Control

In the lottery-choice experiment of Lecture 1b, all participants first made 10 decisions at the **1**× payoff scale, then the same 10 decisions at a high *hypothetical* scale, then at a high *real* scale, and finally again at **1**×. The number of safe choices rises with the real scale.

Why did Harrison et al. (2005) call this design a possible fatal flaw?

a. Because the subjects were students
b. Because payoff scale and order of presentation change together
c. Because the payoffs were too small
d. Because the design is between-subject

PART 2

Expected utility and risk attitudes

Expected utility

Lecture 1b – “Expected payoff”, “Expected utility”, “Relative risk aversion”

Lottery L: x_1 with probability p, x_2 with probability $1-p$.

expected value	$\mathbb{E}[L] = p\,x_1 + (1-p)\,x_2$
expected utility	$\mathbb{E}[u(L)] = p\,u(x_1) + (1-p)\,u(x_2)$
certainty equivalent	$u(CE) = \mathbb{E}[u(L)] \;\Rightarrow\; CE = u^{-1}\big(\mathbb{E}[u(L)]\big)$
risk premium	$RP = \mathbb{E}[L] - CE$
CRRA	$u(x) = x^{1-r}$

u concave $\Leftrightarrow$ risk averse $\Leftrightarrow$ $CE < \mathbb{E}[L]$ $\Leftrightarrow$ $RP > 0$ $\Leftrightarrow$ $0 < r < 1$.

Switching point p^*: solves $\mathbb{E}[u(A)] = \mathbb{E}[u(B)]$. Switching later = more risk averse.

Exercise 2.1

A row of a lottery-choice menu

	payoff with prob. p	payoff with prob. $1-p$
Option A (safe)	€81	€25
Option B (risky)	€121	€1

Set $p = 0.5$.

(a) Compute the expected payoff of each option.
(b) The agent has $u(x) = \sqrt{x}$. Compute the expected utility of each option and the corresponding certainty equivalent. Which option is chosen?

Exercise 2.2

The switching point

Same two options as in Exercise 2.1, but now let p vary as you move down the rows of the menu.

(a) At which p does a *risk-neutral* agent switch from A to B?

(b) At which p does an agent with $u(x) = \sqrt{x}$ switch?

(c) What does the comparison tell you about how to read the data from a Holt–Laury menu?

Exercise 2.3

Certainty equivalent

A lottery pays €90 or €10, each with probability $1/2$. The agent has $u(x) = \sqrt{x}$.

Compute the expected value, the certainty equivalent and the risk premium.

How large is the risk premium as a share of the expected value?

Exercise 2.4

Hypothetical bias

In Holt and Laury (2002) subjects also made the same ten choices knowing that they would *not* be paid for that part. Which statement is correct?

a. Subjects were more risk averse with hypothetical payoffs
b. Subjects were less risk averse with hypothetical payoffs
c. Behaviour was identical in the two conditions
d. Subjects made no mistakes with hypothetical payoffs

PART 3

The Allais paradox and Prospect Theory

The independence axiom

$$A \succsim B \qquad \Longleftrightarrow \qquad \alpha A + (1-\alpha) z \;\succsim\; \alpha B + (1-\alpha) z \qquad \alpha \in (0, 1]$$

With probability $1-\alpha$ you get z *whatever you chose*, with probability α you get the lottery you selected. z is the “irrelevant” common component.

Expected utility is **linear in probabilities**, so diluting both options by the same α rescales the comparison without reversing it:

$$u(x_B) > p\, u(x_A) \qquad \Longrightarrow \qquad \alpha\, u(x_B) > \alpha\, p\, u(x_A) \qquad \text{for every } u$$

The Allais paradox – the two problems

Lecture 2 – "The Allais paradox"; percentages are the modal responses

Problem 1			**Problem 2** = Problem 1 diluted by $\alpha = \frac{1}{4}$		
A:	4,000 with prob. 0.80	20%	A:	4,000 with prob. 0.20	**65%**
B:	3,000 with certainty	**80%**	B:	3,000 with prob. 0.25	35%

Both probabilities are multiplied by $\frac{1}{4}$: $0.80 \to 0.20$, $1 \to 0.25$. The prizes never change.

Most people pick **B** in Problem 1 and **A** in Problem 2. The ranking flips, although the only thing that changed is a common factor.

The Allais paradox – why it is a violation

Normalise $u(0) = 0$. The two modal choices say:

$$\begin{aligned} &\text{B in Problem 1:} && u(3000) > 0.80\,u(4000) \\ &\text{A in Problem 2:} && 0.20\,u(4000) > 0.25\,u(3000) \qquad \Big| \times 4 \\ & && 0.80\,u(4000) > u(3000) \end{aligned}$$

The first and the last line are the **same inequality with opposite signs**. No utility function can satisfy both.

Note what this rules out. The problem is *not* that people are too risk averse: u is arbitrary here, and it cancels out. Curvature of u cannot rescue expected utility, so the fix has to come from outside it, which is what Prospect Theory provides.

Exercise 3.1

Two decision problems

Problem 1: A. win €5,000 with probability 80% B. win €4,000 with certainty

Problem 2: A. win €5,000 with probability 16% B. win €4,000 with probability 20%

The agent has $u(x) = \sqrt{x \cdot 0.001}$.

Compute the expected utility of all four alternatives, and say what this agent chooses in each problem.

Exercise 3.2

What the data say

In the experiments of Kahneman and Tversky the modal respondent chose **B in Problem 1** and **A in Problem 2**.

(a) Is this pattern compatible with expected utility theory?

(b) Which axiom does it violate, and why?

Prospect Theory – the two ingredients

Lecture 2 – "Prospect theory"; Lecture 3a – "probability misperception", "reference dependence"

Probability weighting

$$w(p) = \frac{p^{\omega}}{[p^{\omega} + (1-p)^{\omega}]^{1/\omega}}$$

- $\omega \in (0, 1]$; $\omega = 1$ gives no curvature
- estimates: $\omega \approx 0.7$–0.8
- inverse-S: low p *over*-weighted, high p *under*-weighted

Reference dependence

- gains and losses from a reference point, not final wealth
- loss aversion: $|v(-x)| > v(x)$ for $x > 0$
- reflection effect: risk aversion over gains becomes risk seeking over losses

Exercise 3.3

Two Prospect Theory explanations

Explain, in two or three lines each, how Prospect Theory accounts for the pattern in Exercise 3.2:

(a) through probability misperception;

(b) through reference dependence.

Exercise 3.4

The weighting function

Let

$$w(p) = \frac{p^{\omega}}{[p^{\omega} + (1-p)^{\omega}]^{1/\omega}} \qquad \omega = 0.7$$

Compute $w(0.1)$ and $w(0.9)$, and say in each case whether the probability is over- or underweighted.

The fourfold pattern

	Gains	**Losses**
Low probability	risk seeking *buying a lottery ticket*	risk averse *buying insurance*
High probability	risk averse *settling a lawsuit you expect to win*	risk seeking *gambling to avoid a sure loss*

Probability weighting $\times$ diminishing sensitivity.

Exercise 3.5

The reflection effect

Reflect the two problems around zero:

Problem 3: A. $-4{,}000$ with prob. 80% B. $-3{,}000$ with certainty

Problem 4: A. $-4{,}000$ with prob. 20% B. $-3{,}000$ with prob. 25%

What did Kahneman and Tversky observe, what is the phenomenon called, and how does it differ from loss aversion?

PART 4

Bayes' rule, base rates, representativeness

Bayes' rule

Lecture 3b – "The Bayes' rule", "A counting heuristic"

$$\underbrace{\Pr(A \mid c)}_{\text{posterior}} = \frac{\overbrace{\Pr(c \mid A)}^{\text{likelihood}} \overbrace{\Pr(A)}^{\text{prior}}}{\Pr(c \mid A)\Pr(A) + \Pr(c \mid B)\Pr(B)}$$

Independent draws $S = (c_1, \ldots, c_n)$ with replacement:

$$\Pr(S \mid A) = \prod_{t=1}^{n} \Pr(c_t \mid A)$$

Updating draw by draw (each posterior becomes the next prior) and updating on the whole sequence in one step give the **same** answer.

Two systematic mistakes

Lecture 3b – “Base-rate fallacy”; Lecture 4a – Grether (1980)

- **Base-rate fallacy**: $\Pr(A)$ is neglected, and the answer is given as if only the reliability of the signal mattered.
- **Representativeness**: $\Pr(A \mid c)$ is judged by how much c *resembles* what A typically produces.

Grether (1980), \$15 for a correct prediction and \$5 otherwise:

representativeness and Bayes **agree**	$\approx$ 80% correct
representativeness and Bayes **conflict**	$\approx$ 60% correct

Exercise 4.1

Base rates

A screening test for a rare condition is administered to the whole population.

- The condition affects 2 people in every 1,000.
- Anyone who has the condition tests positive: there are no false negatives.
- 5% of the people who do *not* have the condition also test positive.

A randomly selected person tests positive. What is the probability that this person has the condition?

Exercise 4.2

Two cups – the setting

Two cups contain equally sized Amber (a) and Blue (b) marbles.

Cup A: two Ambers and one Blue Cup B: one Amber and two Blues

A die is thrown to select the cup: if the number drawn is 1 or 2 we use Cup A, otherwise Cup B. The selection is hidden. You then observe draws *with replacement* from the selected cup.

(a) What is the prior probability that Cup A is in use?
(b) The sequence observed is **amber – amber – blue**. Before computing anything: which cup *looks* more likely, and why?

Exercise 4.3

Two cups – updating

Same setting as Exercise 4.2.

Use Bayes' rule to compute $\Pr(A \mid a, a, b)$.

Is Cup A or Cup B more likely to be in use?

Exercise 4.4

One more draw

A fourth draw is made from the same cup, and it is **amber**.

Update the probability that Cup A is in use.

Then check your answer a second way, by applying Bayes' rule to the full four-draw sequence in one step.

Exercise 4.5

Naming the bias

The tendency to make mistakes in the assessment of probability, driven by a rough sense of "similarity" between the observed signal and the underlying event, is called:

a. Loss aversion
b. Representativeness bias
c. Risk aversion
d. Probability matching

Complete the sentence: *in the setting of Exercises 4.2 and 4.3, this tendency would induce one to mistakenly attribute _____, because the observed sample of three draws has the same proportions as _____, and in this sense the sample looks _____.*

PART 5

Probability matching

Probability matching

Lecture 4b – “Probability matching”, “Rational behavior”, “Siegel and Goldstein’s Experiment”

One event occurs with probability $p > 1/2$, the other with $1 - p$; $U_C > U_I$.

predict the more likely event	$pU_C + (1-p)U_I$
predict the less likely one	$pU_I + (1-p)U_C$
difference	$(2p-1)(U_C - U_I) > 0$

Always predict the more likely event, whatever the risk attitude: the two payoffs are the same in both lotteries.

Matching: predict each event with a frequency equal to its frequency of occurrence. Hit rate $p^2 + (1-p)^2 < p$.

Siegel and Goldstein: no-pay / pay / pay-loss. Stronger incentives move behaviour away from matching. Animals are always run with incentives, which is why rats look smarter than humans.

Exercise 5.1

Earnings under the two strategies

In a class experiment on probability matching each correct prediction pays €0.25, and teams make predictions over 20 trials. The more likely event was programmed to occur with probability **0.80**.

(a) How much would a team earn *per trial* by being perfectly rational, once it has learnt which event is more likely?

(b) And a team that follows perfect probability matching?

Exercise 5.2

Generalise

Keep the payoff at €0.25 and let p denote the probability of the more likely event.

Write the per-trial expected earnings under the two strategies, and show that matching is always worse for $p > 1/2$.

At which p is the loss largest?

PART 6

Oral questions

Sample oral questions I

The Allais paradox

1. Illustrate what the Allais paradox is.
2. What is the independence axiom?
3. Explain why the Allais paradox represents a violation of the independence axiom.
4. Illustrate how reference dependence, as assumed by Prospect Theory, can provide an explanation of the Allais paradox.

Risk attitudes and the lottery-choice menu

1. How does expected utility theory represent risk aversion?
2. Describe the Holt and Laury lottery-choice task. What does the switching point tell us?
3. What did Holt and Laury find when they scaled payoffs up, and what changes when the payoffs are hypothetical?

Sample oral questions II

Information and learning

1. What is the base-rate fallacy? Give an example.
2. What is representativeness bias, and how did Grether design an experiment that separates it from Bayesian updating?

Probability matching

1. What is the behavioural bias that goes under the name of “probability matching”?
2. Can you describe the experiment used to study this phenomenon?
3. What is the rational behaviour in this experiment, and what behaviour was observed instead among many respondents?
4. Simple animals like chicks and rats did not fall prey to probability matching, while humans did. Which explanation was given for this surprising result?